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Think Manifolds, Not PDEs

Bayesian Cybersecurity

Where the Modak–Walawalkar Framework sits against the published record

In brief

• **Question:** has anyone already framed physics as a learned manifold of admissible states, built from Bayesian priors, that replaces deriving and solving the PDEs, with one distance serving as both the law and the risk?

• **Finding:** across 12 web searches on 29 October 2026, **no source was found that makes that combined claim**. Every ingredient has prior art, and several neighbouring fields (latent-space reduced-order models, Lagrangian neural networks, manifold-distance anomaly scores) come close. None combines priors as the specification, a Riemannian metric on the learned set, and one distance as law and risk.

• **The tagline:** the exact phrase “Think Manifolds, Not PDEs” returned no matching source. The framework authors' names returned no external hits, which is a sign of indexing, not of absence of the work.

• **What this does not do:** a web search can find prior art; it cannot prove none exists. This is a working search, not a systematic review or a freedom-to-operate opinion. Section 6 lists what a formal search should add before “first” is used without a qualifier.

1 · The claims tested

#	Claim, as the MW paper states it
C1	Physical law is represented as a learned Riemannian manifold of admissible states, not as derived PDEs.
C2	Physics is written as Bayesian priors (Beta, truncated, LogNormal, Bernoulli, Student-t) that act as the specification.
C3	Forecasting is a walk on the manifold, replacing the solve; cost tracks the manifold dimension, not the ambient space.
C4	One distance is both the physical model and the risk measure: Synge's world function on the manifold, a physics-weighted extension off it.
C5	The same engine ports across domains by changing only the priors.

2 · Method

Twelve web searches were run on 29 October 2026, in two groups: eight aimed at the neighbouring fields below, and four on the battery application and the exact tagline. Results are search-engine snippets plus the pages they point to; full texts were not read except where noted. Searches covered: physics-informed VAEs and Riemannian manifolds for battery SoH; edge and CPU battery SoH; learned latent manifolds replacing PDE solves; autoencoder reduced-order models and LaSDI; the geometry of VAE latent spaces; Hamiltonian and Lagrangian neural networks; manifold-distance anomaly scores for cyber-physical systems; neural operators and PINNs; Kennedy–O'Hagan calibration for batteries; and the tagline and author names.

3 · What exists, and how MW differs

Field	Representative work	What it does	How MW differs
Latent-space reduced-order models	LaSDI and variants (GPLaSDI reports up to 100,000× speed-up); VAE-DNN surrogates; physics-constrained latent dynamics, ICLR 2023	Compress PDE solution fields with an autoencoder, identify or learn dynamics in the latent space, skip the full solve at inference.	**Closest to C3.** They need a PDE solver to generate training data and identify latent ODEs. MW specifies the physics as priors and reads forecasts as geodesic walks; no latent ODE is fitted. A speed-up from skipping the solve is therefore not unique to MW.
Lagrangian and Hamiltonian networks	Lagrangian Neural Networks (Cranmer et al.)	Learn a Lagrangian or Hamiltonian from trajectories; equations of motion follow from Euler–Lagrange.	Keeps the 1687–1788 pipeline (Lagrangian, then equations, then solve) and learns its first step. MW drops that pipeline. This is the sharpest contrast with C1.
Geometry of generative models	Latent Space Oddity (Arvanitidis et al., ICLR 2018)	Pull-back Riemannian metric on a VAE latent space; geodesics and distances.	**Source of the metric in C1 and C4**, already cited by the MW paper. No physics priors or domain use in that work.

Field	Representative work	What it does	How MW differs
Manifold-distance anomaly scores	Anomaly scores for generative models; autoencoders for cyber-physical data	Reconstruction error read as squared distance to the learned manifold; limits noted away from the training data.	**Closest to C4's risk side.** MW weights the distance by physics-derived importances and ties it to Synge's world function with a Van Vleck bound. The plain idea of distance-to-manifold as an anomaly score is not new.
Physics-informed battery SoH	PINN for SoH (Nature Commun. 2024); Pace, edge demo; PG-M2TN, edge-friendly	Physics terms in the loss or features; some run on a Raspberry Pi.	None found that uses a VAE manifold for SoH, or CAN-only input at fleet scale. Cheap-hardware SoH is not unique.
Bayesian calibration with a physics prior	GP regression for SoH forecasting (Richardson, Osborne, Howey)	Gaussian processes with degradation-shaped mean functions and credible intervals.	Same Bayesian spirit (the MW paper cites Kennedy–O'Hagan). MW puts the physics in the prior family and a learned manifold in place of the GP.
Neural operators and PINNs	FNO, DeepONet, Raissi et al. (2019)	Faster solves of a PDE that is already derived.	The MW paper's own framing. These make the solve faster; they keep the PDE. (No head-to-head benchmark was found in this search.)
Geometry in PDE theory	Olver, *Equivalence, Invariants and Symmetry*	PDEs studied and solved on manifolds; differential equations as geometric objects.	Different meaning of "manifold". Worth one line in the paper so a reader does not confuse the two.

4 · Reading the result

C2 and C5 (priors as specification, porting): not found in this form. Priors are used in many Bayesian physics models, but not as the primary description of the law with a learned manifold on top, and not carried across eight domains by swapping priors.

C1 and C3 (manifold replaces derive-and-solve): the nearest prior art is latent-space reduced-order modelling. It shows that skipping the solve with a learned low-dimensional space is established. The part not found is deriving nothing at all: no solver for training data, no latent ODE, forecasts as walks on a prior-defined manifold.

C4 (one distance as law and risk): distance-to-manifold anomaly scoring and Riemannian metrics on VAEs both exist. The link to Synge's world function and the physics-weighted off-manifold extension were not found elsewhere. This is a mathematical claim and should be checked by a geometer; the paper already lists the on/off-manifold lemma as a next step.

The tagline: "Think Manifolds, Not PDEs" is available as a phrase. As a slogan it contrasts with Lagrangian neural networks and with neural operators, both of which keep the PDE.

5 · Wording that the search supports

Use	Avoid
"To our knowledge, the first framework in which physics is written as Bayesian priors, learned as a Riemannian manifold, and read with one distance as both law and risk." (already in the paper, Section 4)	"No one else does manifold work." Manifold learning, latent ROMs and Riemannian VAEs are active fields.
"Replace the solve with a search on a learned manifold." Cost grows with manifold dimension, not the ambient space.	"Nobody questioned the pipeline for 339 years." Reduced-order and latent surrogates have questioned the cost of the solve for decades.
"Physics AI without GPUs for batteries."	"No one can replace GPUs." The claim is scoped to this problem.
"Think Manifolds, Not PDEs" as the framework's tagline.	"Thought impossible by all." This cannot be shown from the record.

6 · Limits of this search, and what to add

• **Scope:** one general web search engine, 12 queries, one day. Not covered: Google Scholar and Scopus citation chains, arXiv listing searches by author and by term, patent databases (Espacenet, Google Patents, Indian IPO), and conference proceedings behind paywalls.

• **Not retrieved:** the MW publications listed in the paper (Zenodo and LinkedIn). Search engines did not return them, so they are not independently dated here. A DOI and timestamp per paper (the Zenodo DOI is in the paper) would document priority.

• **Add to the paper's related-work paragraph:** latent-space dynamics identification (LaSDI and SINDy-autoencoder lineage), Lagrangian and Hamiltonian neural networks, and manifold-distance anomaly scoring. Naming them strengthens the "what is new" argument, as Section 4 already says.

• **Before using “first” without a qualifier:** run the formal searches above, with search terms: “physics priors” with “Riemannian VAE”; “learned manifold” with “admissible states”; “replace PDE” with “geodesic”; “world function” with “anomaly”. Ask one outside geometer and one outside battery-modelling researcher to try to name a counterexample.

• **Companion document:** a battery-only version of the framework paper (no other verticals) is a separate piece; this document is the bird's-eye prior-art view across all domains.

Prepared 29 October 2026 for Bayesian Cybersecurity. Search results are third-party snippets and were not independently verified; links were taken from the search output.